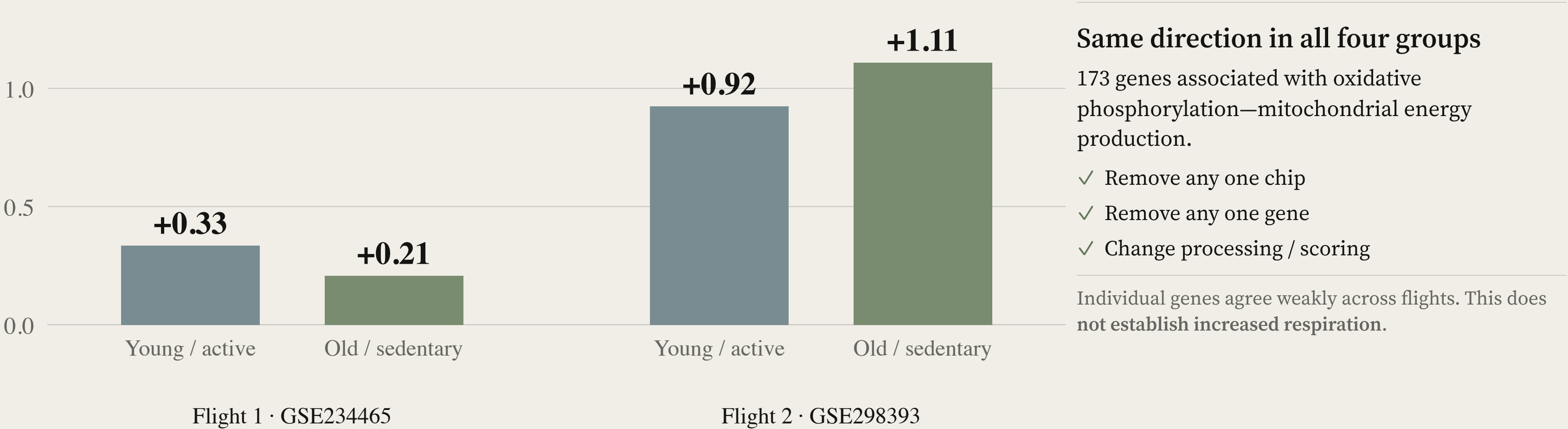


Muscle chips share an average mitochondrial gene-set *increase*

Survey: 10 biological gene sets · 60 primary comparisons · frozen BridgeRNA + expression baseline

Average gene-set score: flight – ground



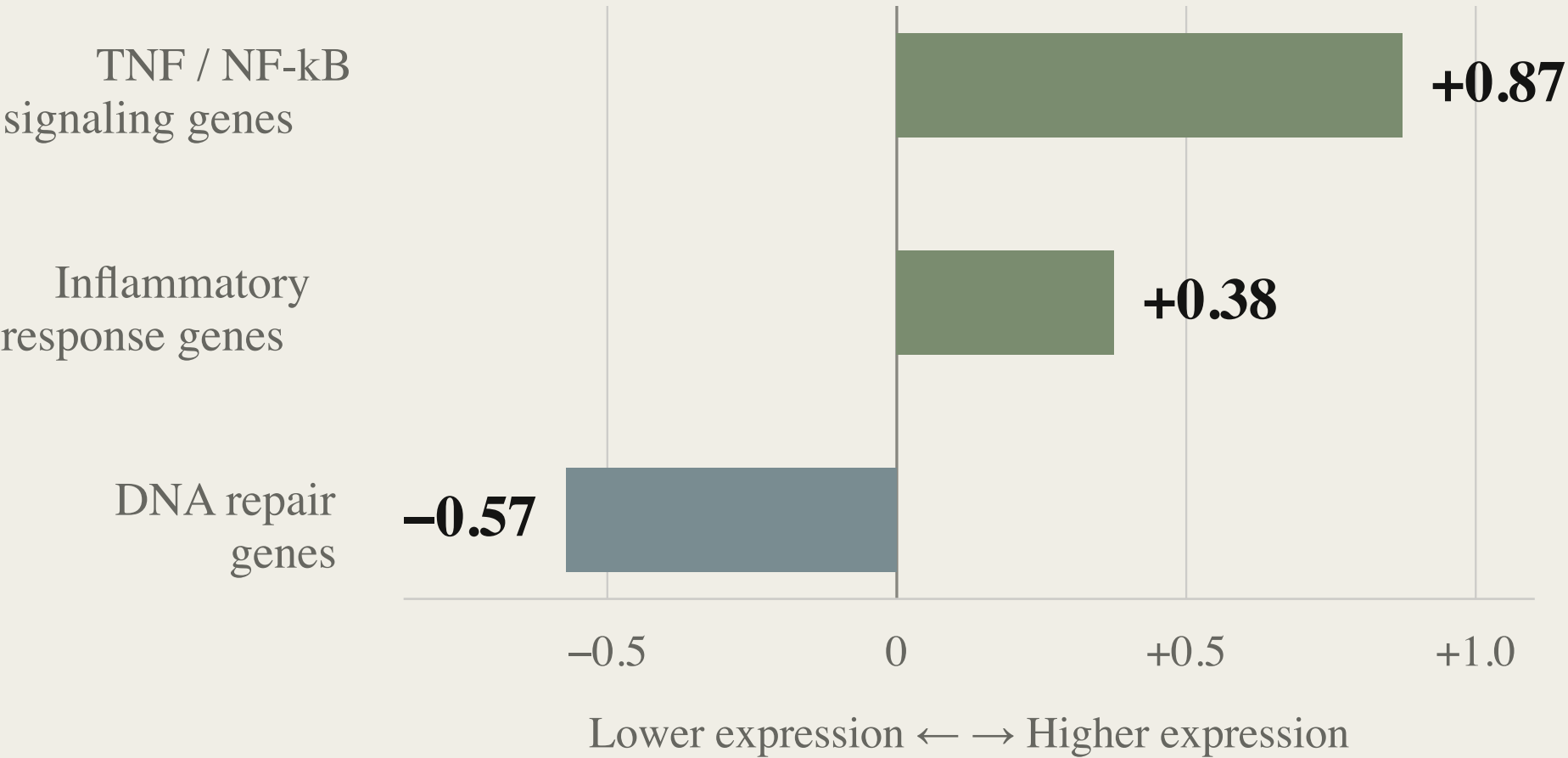
A narrower pattern survives even though the overall expression and model responses differ between flights.

3 chips / condition / pool / flight; shared donor pools. Scores standardized within each comparison; magnitudes are not comparable functional effects.

Onboard 1g controls reveal a *stable* MG63 response

Both groups on the ISS: 3 microgravity cultures ↔ 3 onboard-1g cultures

Average gene-set score: microgravity – onboard 1g



The model retains the response

0.95 **0.83**

BridgeRNA mean Expression

Median split-direction agreement
+1 = same direction · 0 = no alignment

All 9 splits agree in direction for both. Gene-set directions survive sample and source checks.

MG63 is an osteosarcoma-derived cell line. This is a transcriptional lead, not evidence of impaired DNA repair.

Discuss an inflammation / repair-associated response under a more closely matched flight control.

Three highlighted sets from the ten-set panel. Alternate processing: 5 matched samples. Split agreement is not prediction accuracy or proof of model superiority.

Choose one biological question and an *independent* test

Two preliminary leads; each needs a different follow-up.

OPTION A · HUMAN MUSCLE			OPTION B · HUMAN BONE CELLS		
What drives the shared gene-set average?			Does the controlled MG63 pattern recur?		
Audit	→	Member genes, donor pools and culture / collection conditions	Audit	→	Inflammatory and DNA-repair gene-set members
Test	→	Another flight under matched conditions	Test	→	Another onboard-1g study, preferably primary cells
Advance if	→	The direction repeats with interpretable gene-level support	Advance if	→	The expression directions survive an independent comparison

Why controls matter Endothelial mitochondrial gene set: Earth control → stable | Onboard 1g → fragile

Mentor discussion: which question has the clearest biological value and feasible next dataset?

Exploratory associations; no universal flight signature or established pathway function. Choose the follow-up before deeper evaluation or fine-tuning.